Dajia达佳™

DC—4

低频磁疗机使用说明书

LOW FREQUENCY MAGNETIC FIELD THERAPY APPARATUS OPERATION MANUAL

汕头市医用设备厂有限公司

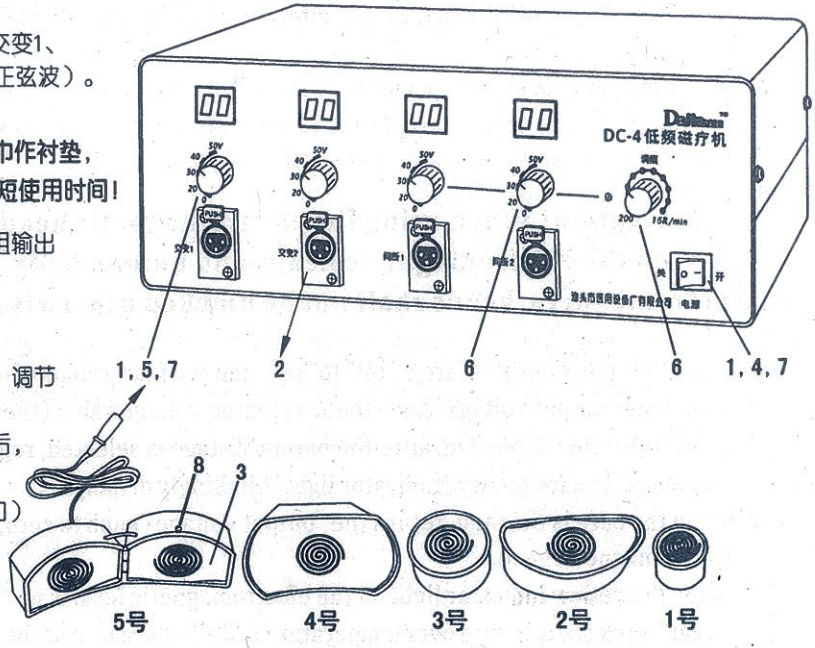
Shantou Medical Equipment Factory Co. ,Ltd.

1. 把交变1、交变2、间断1、间断2四个“输出电压”旋钮回零。电源开关处于“关”断状态。
2. 将选用合适电磁头插入所需波形的插口（交变1、交变2输出正弦波，间断1、间断2输出间断正弦波）。
3. 将电磁头置于患部。

⚠ 注意！电磁头与人体之间应用干燥毛巾作衬垫，并尽量不要堵住电磁头上的散热孔，否则会缩短使用时间！

4. 将“电源开关”由“关”按向“开”，四组输出电压数码管显示“00”字样。
5. 调节“输出电压”旋钮至所需电压值（由数码管显示）。
6. 如需使用“间断”输出，输出电压选定后，调节“调频”旋钮至所需的间断频率。
7. 治疗完毕，先将“输出电压”旋钮调回零后，关断电源。再拆卸电磁头。
8. 电磁头上黄色指示灯亮（使用“交变”插口）或闪烁（使用“间断”插口）时，表示电磁头超温自动保护装置动作，电磁头停止工作，间歇20min后可重新使用。

电磁头每次使用20min是指室温25℃使用“交变”插口且最高输出电压50V时，间歇20min后可重新使用。如果使用“间断”插口或者使用低于50V输出电压时，使用时间可适当延长。



DC-4 低频磁疗机面板结构图及使用提示

Panel structural drawing and usage hints

.. Return the knobs of "alternation 1", "alternation 2", "break 1" and "break 2" to zero. The power switch shall be in the "off" state.

2. Select the suitable electromagnetic heads and plug them in the required wavelike sockets (alternation 1, alternation 2 output sine wave, break 1, and break 2 output sine wave)

3. Place the electromagnetic head on the affected part.

⚠ Caution! When using the electromagnetic heads, dry towels shall be used as pads between the electromagnetic heads and human body, and the thermal via on the electromagnetic heads shall not be blocked otherwise the service life will be reduced.

4. Press the "power switch" from "off" to "on", and the four groups of output voltage nixie tubes show "00".

5. Adjust the "output voltage" knob to the required voltage value (shown on the nixie tubes)

6. If "break" output is needed, after the output voltage is selected, regulate the "FM" knob to the required break frequency. The frequency "indicator light" blinkingly displays.

7. When the cure is finished, return the "output voltage" knob to zero, cut off the power and then disassemble the electromagnetic head.

8. When the yellow indicator light on the electromagnetic head is up (use "alternation" socket) or twinkle (use "break" socket), it is auto overtemperature protection action of the electromagnetic head. The electromagnetic head stops running and can be reused after 20 minutes' interval.

Note: Each use of electromagnetic head for 20 minutes means that under the room temperature of 25℃, when using the "alternation" socket and the highest output voltage is 50V, the electromagnetic head can be reused after 20 minutes' interval. If use the "break" socket or the output voltage is lower than 50V, the usage time can be extended.

DC－4 低频磁疗机使用说明书

医疗器械生产企业许可证：粤食药监械生产许 20010064 号

医疗器械注册证：粤食药监械（准）字 2009 第 2260315 号

产品标准号：YZB/粤 0253－2009

⚠注意！使用本机前，必须详阅使用说明书

1 用途和结构

低频磁疗机是利用低频交变磁场调节人体生物电磁信息，使其恢复正常生理功能，以达到治疗疾病的目的。同时本机对人体辅以电磁震颤，具有按摩功能和热敷作用。

本机对三叉神经痛、前列腺炎、支气管炎、肩周炎、急性扭伤等疾病有较好的辅助治疗作用。实际使用须遵医嘱。参见《实用理疗技术手册》（第二版 2006.10 人民军医出版社）。

本机由台式主机、电磁头等组成。主机面板上有四路输出插口（交变 1、交变 2 输出正弦波，间断 1、间断 2 输出间断正弦波）。四个调节旋钮，分别用于调节对应插口的输出电压，并由四组数码管显示。"调频"旋钮用于调节"间断"输出频率（16 次/min～200 次/min），并由指示灯闪烁显示。

主机外形尺寸为 400mm × 240mm × 150mm，重量为 12kg，电磁头全套重量为

10kg。

2 要求

2.1 本机应符合 YZB/粤 0253－2009《低频磁疗机》的要求。安全性能应符合 GB9706.1－2007《医用电气设备 第一部分:安全通用要求》的要求。按医用电气设备电击防护分类为Ⅰ类设备、有保护接地符号“⏚”,按防电击的程度为 B 型应用部分、符号“⚟”。

2.2 正常工作条件

a) 环境温度为 5℃～40℃。

b) 相对湿度≤80%。

c) 电源额定电压和频率:a.c. 220V,50Hz,输入功率 500VA。

d) 工作制:主机连续工作时间为 4h;电磁头使用 20min、间歇 20min。

2.3 输出电压(在交变插口测试,输入电压 a.c.220V 时)

分 20V、30V、40V、50V 四档,每档误差±10%。

2.4 输出波形(分四路输出)

a)正弦波二路(“交变 1”插口和“交变 2”插口)

b)间断正弦波二路(“间断 1”插口和“间断 2”插口),间断频率不窄于 16 次/min～200 次/min,连续可调。

2.5 电磁头的性能要求按表 1 的规定(使用交变插口,输出调至最大)。

表 1　电磁头的性能要求

电磁头编号	输出磁通量密度 mT(Gs 高斯)	输出振动幅度 mm
1	＞ 15(150Gs)	(加载 1.5kg,法码 Φ50mm) ＞ 0.006
2	＞ 20(200Gs)	(加载 1.5kg,法码 Φ50mm) ＞ 0.008
3	＞ 20(200Gs)	(加载 2kg,法码 Φ70mm) ＞ 0.008
4	＞ 20(200Gs)	(加载 2kg,法码 Φ70mm) ＞ 0.008
5	＞ 20(200Gs)	(加载 2kg,法码 Φ70mm) ＞ 0.008

3　使用方法

⚠注意！使用本机前，必须详细阅读禁忌症、注意事项、警示及提示性说明！

使用本机时，请参照面板结构图，并按使用提示操作。

4　禁忌症、注意事项、警示及提示性说明

4.1 禁忌症(参见《实用理疗技术手册》第二版 2006.10)

恶性肿瘤、高热、急性化脓性炎症、出血倾向、活动性结核、妊娠期妇女、局部金属异物、心脏起搏器局部及其邻近、心力衰竭、无自主行为意识者禁用。

4.2 不得与高频手术设备、短波及微波设备同时使用。

4.3 不得在高温、高湿、易燃、失衡、烟尘过量、电磁幅射的场所使用本机。

4.4 操作本机必须是经培训的医护人员。

4.5 保证本电疗机的网电源及保护接地环境安全稳定。(本机电源插头上有符号⏚的为保护接地点)

4.6 使用电磁头时,电磁头与人体之间应用干燥毛巾作衬垫,并尽量不要堵住电磁头上的散热孔,否则会缩短使用时间。

4.7 熔断器和其他部件的更换

本机无客户可自行维修的组件,发生故障时请勿自行拆卸,应请专业人员维修。网电源电路中的保险丝管为 UTE・T 2A L250V,维修更换时应用相同型号规格的保险丝管。

4.8 使用电疗机应先了解各功能开关的作用,使用时应准确选择各开关位置,切勿拨错。

4.9 环境保护

废弃物、残渣等的处理应按当地有关部门的规定处置,本产品使用寿命到期后或维修换下的配件请妥善处理,以免污染环境。

4.10 制造商责任

只有在以下几种情况下,制造商、装备者、安装者或进口商才认为对设备的安全性、可靠性和性能方面受到的影响负有责任:

——装配、增设、调试、改动或维修都是由他认可的人员进行的;

——有关房间内的电气设施是符合有关要求的,以及

——设备是按使用说明书要求使用的。

5 成套附件

5.1 电磁头一套分:1 号、2 号、4 号电磁头各一只,3 号、5 号电磁头各二只。

5.2 电源线一条。

5.3 合格证一份。

5.4 使用说明书一份。

注:本使用说明书不附电原理图和元器件清单,如用户因维修需要,可向供方索取。本使用说明书的内容已包含技术说明书内容。

6 维护和保养方法

6.1 每年应对产品进行一次检验,确保各项功能正常运作。

6.2 常见故障、故障原因和解决方法见表 2。

表 2 常见故障、故障原因和解决方法

常见故障	故障原因	解决方法
开机后无输出	电源插头未插好	重新插好电源插头
	电源保险丝管 UTE · T2AL250V 断路	更换电源保险丝管
	输出插口与电磁头之间未插好	检查输出插口与电磁头应插紧
电磁头振动幅度太小	输出电压偏低	调高输出电压
电磁头上的黄色指示灯亮并停止工作	并非故障,是使用时间过长,电磁头超温,超温自动保护装置动作	关机,让它自然冷却 20min 后再使用。

6.3 出现未能解决的故障时,应请专业人员检修。

6.4 低频磁疗机应保持清洁及卫生,请用柔软的抹布和洗涤液进行清洁处理,每次清洁

后应晾干才使用。

6.5 与患者接触部件的清洗、消毒和灭菌

电磁头可用柔软的抹布和洗涤液对其塑料外壳进行清洁处理，用医用酒精消毒。电磁头与人体之间应用干燥毛巾作衬垫，更换患者时应同时更换经消毒处理的干燥毛巾，避免交叉感染。

6.6 运输和贮存的环境条件

a)产品允许使用一般运输工具，但须防止运输过程中的剧烈冲击、震动及雨雪淋溅。运输要求按订货合同规定。

b)包装后的产品应贮存在温度－10℃～40℃、相对湿度不大于80%、无腐蚀气体和通风良好的室内。

6.7 本产品保修期限为一年。

7 产品及包装图形、符号说明

7.1 ⏚ Ⅰ类设备、有保护接地

7.2 B型应用部分

7.3 ⚠ 注意！参考随附文件

7.4 ○ 电源断开

7.5 | 电源接通

7.6 易碎、小心轻放

7.7 向上

7.8 保持干燥

参考文献：《实用物理治疗手册》(第二版2006.10人民军医出版社)。(www.pmmp.com.cn)

2012年1月印制

Operation manual of DC－4
Low Frequency Magnetic Field Therapy Apparatus

Medical Device Manufacturing Enterprise License: YSYJXSCX No.20010064
Medical instrument registration certificate: YSYJX (Z) Z 2009 No. 2260315
Product Standard No.: YZB/Y 0253－2009

⚠ Caution! Before using this machine, you must read the operation manual carefully.

1. Application and structure

The low frequency magnetic field therapy apparatus regulates bioelectromagnetic information of human body to restore its normal physiological function by using low－frequency alternating magnetic field for the purpose of curing the disease. And the machine can also aid human body with electromagnetic tremor, having functions of massage and hot compression.

This machine offers good adjutant therapy to such diseases as trigeminal neuralgia, prostatitis, bronchitis, periarthritis humeroscapularis and acute sprains. It shall be used as directed by a physician. Refer to the Practical Physiotherapy Technical Manual (Second Edition, October, 2006, People's Medical Officer Press).

This machine is composed of principal machine and electromagnetic heads and so on. There are four output sockets (alternation 1, alternation 2 output sine wave, break 1, and break 2 output sine wave) on the panel of the principal machine. Four adjusting knobs are used for regulating the output voltages of the corresponding sockets respectively, which are displayed by four groups of nixie tubes. The "FM" knob is used for regulating the "break"

output frequency (16 times/min～200times /min), which is blinkingly displayed by the indicator lights.

The external dimension of the principal machine is 400mm×240mm×150mm and its weight is 12kg, of which, the whole set of electromagnetic heads weight 10kg.

2. Requirements

2.1 This machine complies with the requirements of YZB/Y 0253－2009 LOW FREQUENCY MAGNETIC FIELD THERAPY APPARATUS. Its safety performance shall comply with the requirements of GB9706.1－2007 Electrical Equipment for Medical Use Part 1: General safety requirements. With regard to protection against electric shock of electrical equipment for medical use, it is classified as class Ⅰ equipment with protective earthing sign "⏚"; with regard to the degree of protection against electric shock, it is classified as type B application part with the sign of"⚠".

2.2 Nominal operating conditions

a) Ambient temperature is 5℃～40℃.

b) Relative humidity≤80%.

c) Rated supply voltage and frequency: a. c. 220V,50Hz, input power 500VA.

d) Character of service: the continuous operating time of the principal machine is 4h; the electromagnetic heads serve 20 min and then stop for 20min.

2.3 Output voltages (tested at the alternating sockets and when the input voltage is a. c. 220V) are divided into four grades: 20V, 30V, 40V and 50V, the error of each grade ±10%.

2.4 Output waveform (four outputs)

a) Two circuits of sine－wave ("alternation 1" socket and "alternation 2" socket)

b) Two circuits of breaking sine－wave ("break 1" socket and "break 2" socket), the break frequency shall not be narrow than 16times/min～200times/min and shall be continuously adjustable.

2.5 The performance requirements of the electro-

magnetic heads shall comply with the regulations in the Table 1 (use alternating socket and regulate the output to maximum.)

Table 1 The performance requirements of the electromagnetic heads

Electromagnetic heads No.	The output magnetic flux density (mT) (Gs Gauss)	Output oscillation amplitude (mm)
1	> 15(150Gs)	(Load 1.5kg , weight Φ50mm) > 0.006
2	> 20(200Gs)	(Load 1.5kg , weight Φ50mm) > 0.008
3	> 20(200Gs)	(Load 2kg , weight Φ70mm) > 0.008
4	> 20(200Gs)	(Load 2kg , weight Φ70mm) > 0.008
5	> 20(200Gs)	(Load 2kg , weight Φ70mm) > 0.008

3. Use methods

⚠ Caution! Before using this machine, you must read the Contraindications, Precautions, Warnings and Instructional Statements carefully.

When using this machine, please refer to the panel structural drawing and operate in accordance with the use hints.

4. Contraindications, Precautions, Warnings and Instructional Statements

4.1 Contraindications (see the Practical Physiotherapy Technical Manual Second Edition, October, 2006)

Patients with malignant tumor, hyperpyrexia, acute suppurative inflammation, hemorrhagic tendency, active tuberculosis, women in gestational period, local metal foreign body, cardiac pacemaker local and its surrounding heart failure or without awareness of voluntary behavior are forbidden to use it.

4.2 It can not be used with high-frequency surgical apparatus, short-wave and micro-wave apparatus at the same time.

4.3 It shall not be used in the places of high temperature, high humidity, unbalance, excess smoke dust and electromagnetic radiation or the explosive places.

4.4 This machine must be operated by the trained medical care personnel.

4.5 Ensure that the network source and protective earthing environment are safe and stable. (Those with signs ⏚ on the power plug of this machine are protective earthing points.)

4.6 When using the electromagnetic heads, dry towels shall be used as pads between the electromagnetic heads and human body, and the thermal via on the electromagnetic heads shall not be blocked otherwise the service life will be reduced.

4.7 Replacement of the fuse and other components

There is no user serviceable parts in this machine and do not attempt to disassemble it when a fault occurs. It shall be maintained by professional staff. The cartridge fuses in the network source circuit are UTE · T 2A L250V. The cartridge fuses with the same models shall be used in the maintenance and replacement.

4.8 When using the electric therapy apparatus, one shall learn the function of each switch and can accurately choose the position of each switch. Be sure not toggle the wrong switch.

4.9 Environmental Protection

The disposal of waste and residue and so on

shall be in accordance with the provisions of local related departments. The machines, the service life of which has expired, and the accessories replaced in the maintenance shall be treated well to avoid polluting the environment.

4.10 Manufacturer's liability

The manufacturers, assemblers, installers or importers shall be responsible for the influence the equipments affected in respects of safety, reliability and performance only under the following situations:

—— The assembling, adding, debugging, change and maintenance are made by the personnel approved by the manufacturer.

——The relevant electric installations in the room are subject to the relevant requirements, and

—— The equipments are used according to the operation manual.

5. Adapter kit

5.1 One set of electromagnetic heads including: one for each of No.1, 2, 4 electromagnetic heads; two for each of No. 3 and No.5 electromagnetic heads.

5.2 One power cord

5.3 One certificate of compliance

5.4 One operation manual

Note: This operation manual not include electrical schematic diagram and components list. If users need them for maintenance, they can demand from the supplier. The contents of this manual have included the contents of engineering instructions.

6. Maintenance and care methods

6.1 The product shall be inspected once per year to ensure the normal running of each function.

6.2 Common fault, fault cause and methods of resolution see Table 2.

Table 2 Common fault, fault cause and methods of resolution

Common faults	Fault causes	Methods of resolution
No output after power on	The power plug is not plugged well.	Re—plug the power plug.
	The power supply fuse UTE · T2AL250V open	Replace the power supply fuse
	The electromagnetic head is not plugged well in the output socket.	Check whether the electromagnetic head is plugged well in the output socket.
The oscillation amplitude of electromagnetic head is too small.	The output voltage is low.	Heighten the output voltage.
The yellow indicator light on the electromagnetic head lights up and stops working.	It is not fault. It is the automatic over-temperature protection action when the electromagnetic head is over temperature due to long time service.	Close down, and reuse it after 20 minutes natural cooling.

6.3 When unresolved faults occur, shall ask professional personnel to overhaul.

6.4 The low frequency magnetic field therapy apparatus shall be kept clean and sanitary. The cleaning process shall be made with soft cleaning rag and cleaning solution. After cleaning, it shall be

used after been dried.

6.5 The cleaning, disinfection and sterilization for the components connecting with the patients.

The plastic shell of the electromagnetic head can be cleaned with soft cleaning rag and cleaning solution and be disinfected with medicinal alcohol. A dry towel shall be used as pad between the electromagnetic head and human body. When change patient, the disinfected dry towel shall also be changed to avoid cross contamination.

6.6 The environment conditions of transportation and storage

a) The common means of transport are allowed for use, but radical impact, vibration and exposure to rain or snow in the transportation process shall be avoided. The transportation requirements shall be in accordance with the regulations in the order contract.

b) The packaged products shall be stored in the well－ventilated room with the temperature at －10℃ ～40℃, relative humidity not more than 80% and without corrosive gas.

6.7 The warranty period of this product is one year.

7. Product and package drawing and symbol description

7.1 ⏚ Class Ⅰ equipment, with protective earthing

7.2 type B application part

7.3 ⚠ Caution! Refer to the auxiliary instruments

7.4 ○ Power off

7.5 | Power on

7.6 Fragile, handle with care

7.7 Upwards

7.8 Keep dry

References: Practical Physiotherapy Technical Manual (Second Edition, October, 2006, People's Medical Officer Press). (www.pmmp.com.cn)

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